

Figure 4: Enabling on-the-fly projection and choosing the common projection

and all the subsequent layers will be using in common. Let us choose WGS 84. Click OK.

You can immediately see the result—both the layers are overlaid on top of each other in the same view. In case you can't see the result immediately, then right click on one of the layers in the table of contents, and select **Zoom to layer extent**.

Assigning a projection to a raster layer that has no projection defined

For this, I used the freely available Miniscale raster layer data for the UK. This layer was downloaded from the UK's OS open data initiative (<https://www.ordnancesurvey.co.uk/opendatadownload/products.html>).

We will also be using the freely available vector data downloaded from Natural Earth (<http://www.naturalearthdata.com/>), particularly the `ne_10m_admin_0_countries` layer.

After downloading the data, the Zip file is extracted. The folder we used was the `RGB_TIF_COMPRESSED` folder. This folder contains the Tiff files (the plain Tiff files, not the Geotiff files), which means that it has no projection

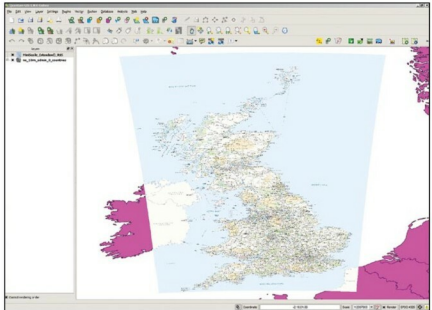


Figure 5: It is clearly visible that the raster layer is correctly overlaid on top of the vector layer

assigned to it. The projection information is usually stored in the `.prj` file format.

Also, under the `RGB_TIF_COMPRESSED/Data` folder, we can see a folder named `world files`. Typically, to open a GIS file in GIS software, we need two types of information, which are:

- The extent of the image
- The projection file (`.prj`)

The `world file` is a standard file format introduced by ESRI. This file stores the information on the extent of the raster image. A `.prj` file stores the projection information of the layer. GIS software can render a GIS file properly if these two files are stored in the same directory as the image file and also have the same file name as the image file.

Click on **Layer Menu > Add raster**

layer. Select `MiniScale_(standard)_R15.tif` file and click on **Open**.

Select the CRS for the layer. The documentation for the raster layer can be downloaded from the link <http://www.ordnancesurvey.co.uk/oswebsite/docs/user-guides/os-opendata-getting-started-guides.pdf>. After going through the documentation, it can be determined that the layer is geo-referenced in the British national grid projection. Select it from the list and click OK.

After the layer is loaded, right-click on the layer and select **Zoom to extent**.

Once zoomed in to the layer, we can see that the newly added raster layer is overlaid perfectly on top of the vector layer. This is how a projection is assigned to a new layer. **END**

By: Abhijith Sastry N S

The author has completed his masters in geoinformatics and has more than two years of work experience in the field of GIS. He is an independent researcher, who primarily works in the field of GIS and its applications. He can be contacted at abhijithsastry@gmail.com.

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